Bioinformatics Practical Assignment
Title – OPTION A .BRCA1 -BREAST Cancer Gene

Retrieval and Analysis of the BRCA1 Gene Sequence Using Bioconductor Packages in R
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Submission Date: __10 August__________________
Introduction
BRCA1 (Breast Cancer 1) is a well-known tumour suppressor gene involved in repairing damaged DNA through the homologous recombination pathway. Mutations in this gene are strongly associated with hereditary breast and ovarian cancers. In this practical, the BRCA1 reference sequence was downloaded from the NCBI database and analysed using several Bioconductor packages in R, including Biostrings, GenomicRanges, AnnotationDbi, and rentrez.

Aim

The objectives of this practical were to:

Download the BRCA1 nucleotide sequence from the NCBI database.
Save the sequence as a FASTA file.
Determine the sequence length.
Extract the first 30 nucleotides.
Count the occurrence of the "ATG" motif.
Generate the reverse complement of the first 30 bases.
Create a genomic range object for BRCA1.
Retrieve the Entrez Gene ID and official gene name.
Software and Packages Used
R (Version 4.6.1)
RStudio
Biostrings
GenomicRanges
AnnotationDbi
org.Hs.eg.db
rentrez
Practical Work
Loading Required Packages

```
# Install packages (Run only once)

if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install(c(
  "Biostrings",
  "GenomicRanges",
  "AnnotationDbi",
  "org.Hs.eg.db"
```

```r
))

install.packages("rentrez")

# Load libraries

library(Biostrings)
library(GenomicRanges)
library(AnnotationDbi)
library(org.Hs.eg.db)
library(rentrez)
```
Question 1

Q1. Download the BRCA1 sequence using entrez_fetch() and save it as "brca1.fasta". How many bases long is the sequence? (Use width())

R Code

```r
# Download BRCA1 sequence

brca1_fasta <- entrez_fetch(
  db = "nuccore",
  id = "NM_007294",
  rettype = "fasta",
  retmode = "text"
)

# Save FASTA file

writeLines(brca1_fasta, "brca1.fasta")

# Read FASTA

brca1 <- readDNAStringSet("brca1.fasta")

brca1_seq <- brca1[[1]]

# Determine sequence length

sequence_length <- tryCatch(
  width(brca1_seq),
  error=function(e){
    nchar(as.character(brca1_seq))
  }
)

cat("Sequence Length =", sequence_length, "bases\n")
```
Output

```
Sequence Length = 7088 bases
```
Observation

The downloaded BRCA1 reference sequence contains 7,088 nucleotides.

Question 2

Q2. What are the first 30 bases of the sequence? (Use subseq())

R Code

```
first30 <- subseq(brca1_seq,
          start = 1,
          end = 30)

print(first30)
```

Output

30-letter DNAString object

GCTGAGACTTCCTGGACGGGGGACAGGCTG

Observation

The first thirty nucleotides of the BRCA1 sequence were successfully extracted.

Question 3

Q3. How many times does the letter pattern "ATG" appear in the sequence? (Use countPattern())

R Code

```
ATG_number <- countPattern("ATG", brca1_seq)

print(ATG_number)
```

Output

140

Observation

The nucleotide pattern ATG appears 140 times in the BRCA1 reference sequence.

Question 4

Q4. What is the reverse complement of the first 30 bases?

R Code

```
reverse_first30 <- reverseComplement(first30)

print(reverse_first30)
```

Output

30-letter DNAString object

CAGCCTGTCCCCCGTCCAGGAAGTCTCAGC

Observation

The reverse complement sequence was successfully generated using the reverseComplement() function.

Question 5
Q5. Create a GRanges object for BRCA1 using the table above. What is its width (end − start)?
R Code
```
brca1_range <- GRanges(
  seqnames = "chr17",
  ranges = IRanges(
    start = 43044295,
    end = 43125483
  ),
  strand = "-"
)

gene_width <- width(brca1_range)

print(brca1_range)

gene_width
```
Output
81189
Observation

The genomic coordinates indicate that BRCA1 spans 81,189 base pairs.

Question 6
Q6. Using AnnotationDbi and org.Hs.eg.db, what is the ENTREZ ID and full gene name for "BRCA1"?
R Code
```
gene_information <- select(
  org.Hs.eg.db,
  keys = "BRCA1",
  keytype = "SYMBOL",
  columns = c("ENTREZID","GENENAME")
)

print(gene_information)
```
Output

SYMBOL	ENTREZID	GENENAME
BRCA1	672	BRCA1 DNA repair associated

Observation

The annotation database confirmed the official Entrez Gene ID and gene name for BRCA1.

Question 7
Q7. In 2–3 sentences, summarize what you learned about BRCA1's sequence and location.
Answer

The BRCA1 gene is located on chromosome 17 (GRCh38) on the negative DNA strand and spans approximately 81,189 base pairs. During this practical, I retrieved the BRCA1 reference sequence from the NCBI database and analysed it using Bioconductor packages in R. The analysis showed that the sequence is 7,088 bases long, contains 140 occurrences of the "ATG" motif, and its annotation corresponds to Entrez Gene ID 672 with the official name BRCA1 DNA repair associated.

Final Answers
Question	Answer
Q1. Download the BRCA1 sequence using entrez_fetch() and save it as brca1.fasta. How many bases long is the sequence?='	7,088 bases"
Q2. What are the first 30 bases of the sequence?='
	GCTGAGACTTCCTGGACGGGGGACAGGCTG"
Q3. How many times does the pattern "ATG" appear?=	140
Q4. What is the reverse complement of the first 30 bases?=
	CAGCCTGTCCCCCGTCCAGGAAGTCTCAGC
Q5. What is the width of the BRCA1 GRanges object?=	81,189 bp
Q6. What is the ENTREZ ID and full gene name?	Entrez ID: 672; Gene Name: BRCA1 DNA repair associated
Q7. Summary	BRCA1 is located on chromosome 17 on the negative strand. The downloaded reference sequence is 7,088 bases long, spans 81,189 bp in the genome, contains 140 "ATG" motifs, and corresponds to Entrez Gene ID 672.
Conclusion

This practical demonstrated how Bioconductor packages can be integrated to retrieve and analyse biological sequence data in R. The workflow included sequence retrieval from NCBI, sequence manipulation, motif analysis, genomic coordinate representation, and gene annotation. These tools provide an efficient approach for performing fundamental bioinformatics analyses and understanding the characteristics of important human genes such as BRCA1.

OptionC.Bioinformatics Practical Assignment

Title

Retrieval and Analysis of the EGFR Lung cancer\Targeted Therapy Gene Sequence Using Bioconductor Packages in R

Student Name: TUBA DURDANA________________________

Submission Date: _____________________

Introduction

The Epidermal Growth Factor Receptor (**EGFR**) gene is a protein-coding gene located on chromosome 7. It plays a major role in regulating cell growth, proliferation, and survival through signal transduction pathways. Mutations in EGFR are frequently associated with non-small cell lung cancer (NSCLC) and are important targets for targeted cancer therapies. In this practical, the EGFR reference sequence was retrieved from the NCBI database and analysed using Bioconductor packages in R.

Aim

The objectives of this practical were to:

* Download the EGFR nucleotide sequence from NCBI.
* Save the sequence in FASTA format.
* Determine the sequence length.
* Extract the first 30 nucleotides.
* Count the occurrence of the "ATG" motif.
* Generate the reverse complement of the first 30 nucleotides.
* Create a GRanges object for EGFR.
* Retrieve the Entrez Gene ID and official gene name.

Software and Packages

* R (Version 4.6.1)
* RStudio
* Biostrings
* GenomicRanges
* AnnotationDbi
* org.Hs.eg.db
* rentrez

R Script

```r
################################################################
# Bioinformatics Assignment
# Gene: EGFR
################################################################
```

```
###############################
# Install Packages (Run Once)
###############################

if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install(c(
  "Biostrings",
  "GenomicRanges",
  "AnnotationDbi",
  "org.Hs.eg.db"
))

install.packages("rentrez")

###############################
# Load Libraries
###############################

library(Biostrings)
library(GenomicRanges)
library(AnnotationDbi)
library(org.Hs.eg.db)
library(rentrez)

################################################################
# Q1 Download EGFR Sequence
################################################################

egfr_fasta <- entrez_fetch(
  db = "nuccore",
  id = "NM_005228",
  rettype = "fasta",
  retmode = "text"
)

writeLines(egfr_fasta, "egfr.fasta")

egfr <- readDNAStringSet("egfr.fasta")
egfr_seq <- egfr[[1]]

################################################################
# Q1 Sequence Length
################################################################
```

```
sequence_length <- tryCatch(
  width(egfr_seq),
  error = function(e){
    nchar(as.character(egfr_seq))
  }
)

cat("Sequence Length =", sequence_length, "bases\n")

#################################################################
# Q2 First 30 Bases
#################################################################

first30 <- subseq(egfr_seq, start = 1, end = 30)
print(first30)

#################################################################
# Q3 Count ATG
#################################################################

ATG_number <- countPattern("ATG", egfr_seq)
print(ATG_number)

#################################################################
# Q4 Reverse Complement
#################################################################

reverse_first30 <- reverseComplement(first30)
print(reverse_first30)

#################################################################
# Q5 GRanges Object
#################################################################

egfr_range <- GRanges(
  seqnames = "chr7",
  ranges = IRanges(
    start = 55019017,
    end = 55211628
  ),
  strand = "+"
)

gene_width <- width(egfr_range)

print(egfr_range)
```

```r
cat("Gene Width =", gene_width, "bp\n")

##################################################################
# Q6 Gene Annotation
##################################################################

gene_information <- select(
  org.Hs.eg.db,
  keys = "EGFR",
  keytype = "SYMBOL",
  columns = c("ENTREZID","GENENAME")
)

print(gene_information)
```

  # Question 1

  ### **Q1. Download the EGFR sequence using `entrez_fetch()` and save it as `egfr.fasta`. How many bases long is the sequence?**

  ### Result

  The EGFR reference sequence was successfully downloaded from the NCBI database and saved as **egfr.fasta**.

Answer: **9,905 bases**

  # Question 2

  ### **Q2. What are the first 30 bases of the sequence?**

  ### Result

  The first 30 nucleotides obtained using the `subseq()` function are:

  AGACGTCCGGGCAGCCCCCGGCGCAGCGCG

  # Question 3

  ### **Q3. How many times does the pattern "ATG" appear in the sequence?**

Result

Using the `countPattern()` function, the motif **"ATG"** was found:

160 times

Question 4

Q4. What is the reverse complement of the first 30 bases?

Result

The reverse complement generated using `reverseComplement()` is:

CGCGCTGCGCCGGGGGCTGCCCGGACGTCT

Question 5

Q5. Create a `GRanges` object for EGFR using the table above. What is its width (end − start)?

Result

A `GRanges` object was created using the genomic coordinates:

* Chromosome: **chr7**
* Start Position: **55,019,017**
* End Position: **55,211,628**
* Strand: **+**

Using the `width
()` function, the genomic width is:

192,612 bp

> **Note:** `width()` in GRanges uses inclusive genomic coordinates (`end − start + 1`), therefore the returned width is **192,612 bp**.

Question 6

Q6. Using AnnotationDbi and org.Hs.eg.db, what is the ENTREZ ID and full gene name for "EGFR"?

Result

Gene Symbol	Entrez ID	Official Gene Name
EGFR	**1956**	epidermal growth factor receptor

Question 7

Summary

The EGFR gene is located on chromosome 7 (GRCh38) on the positive DNA strand and spans **192,612 base pairs**. In this practical, I successfully retrieved the EGFR reference sequence from the NCBI database and analysed it using Bioconductor packages in R. The sequence was **9,905 bases** long, contained **160 occurrences** of the **ATG** motif, and its annotation confirmed Entrez Gene ID **1956** with the official gene name **epidermal growth factor receptor**.

Final Answers

Question	Answer
Q1. Download the EGFR sequence using `entrez_fetch()` and save it as `egfr.fasta`. How many bases long is the sequence?	**9,905 bases**
Q2. What are the first 30 bases of the sequence?	**AGACGTCCGGGCAGCCCCCGGCGCAGCGCG**
Q3. How many times does the pattern "ATG" appear in the sequence?	**160**
Q4. What is the reverse complement of the first 30 bases?	**CGCGCTGCGCCGGGGGCTGCCCGGACGTCT**
Q5. What is the width of the EGFR `GRanges` object?	**192,612 bp**

| **Q6. What is the ENTREZ ID and full gene name for EGFR?**
| **Entrez ID:** 1956; **Gene Name:** epidermal growth factor receptor
|
| **Q7. Summary** | EGFR is located on chromosome 7 on the positive strand. The downloaded reference sequence is **9,905 bases** long, spans **192,612 bp** in the genome, contains **160** occurrences of the **ATG** motif, and corresponds to Entrez Gene ID **1956** with the official gene name **epidermal growth factor receptor**. |

Conclusion

This practical demonstrated how Bioconductor packages can be used together to retrieve, analyse, and annotate biological sequence data in R. The workflow included downloading a reference sequence from NCBI, performing sequence analysis with Biostrings, representing genomic coordinates using GenomicRanges, and obtaining gene annotation through AnnotationDbi. These tools provide a practical and reproducible approach for analysing biologically important genes such as EGFR.

END OF THE PROJECT.
THANKYOU.